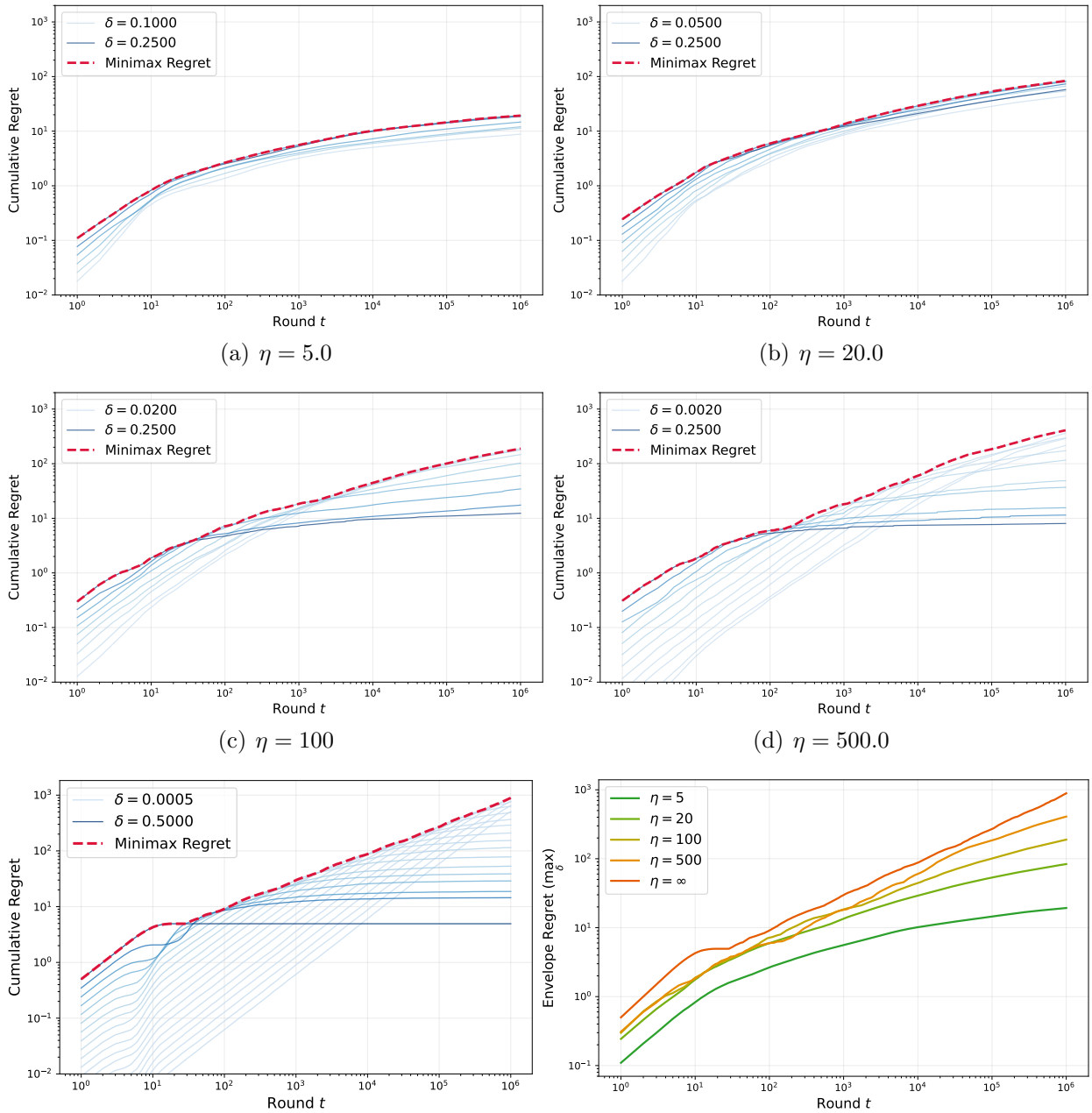
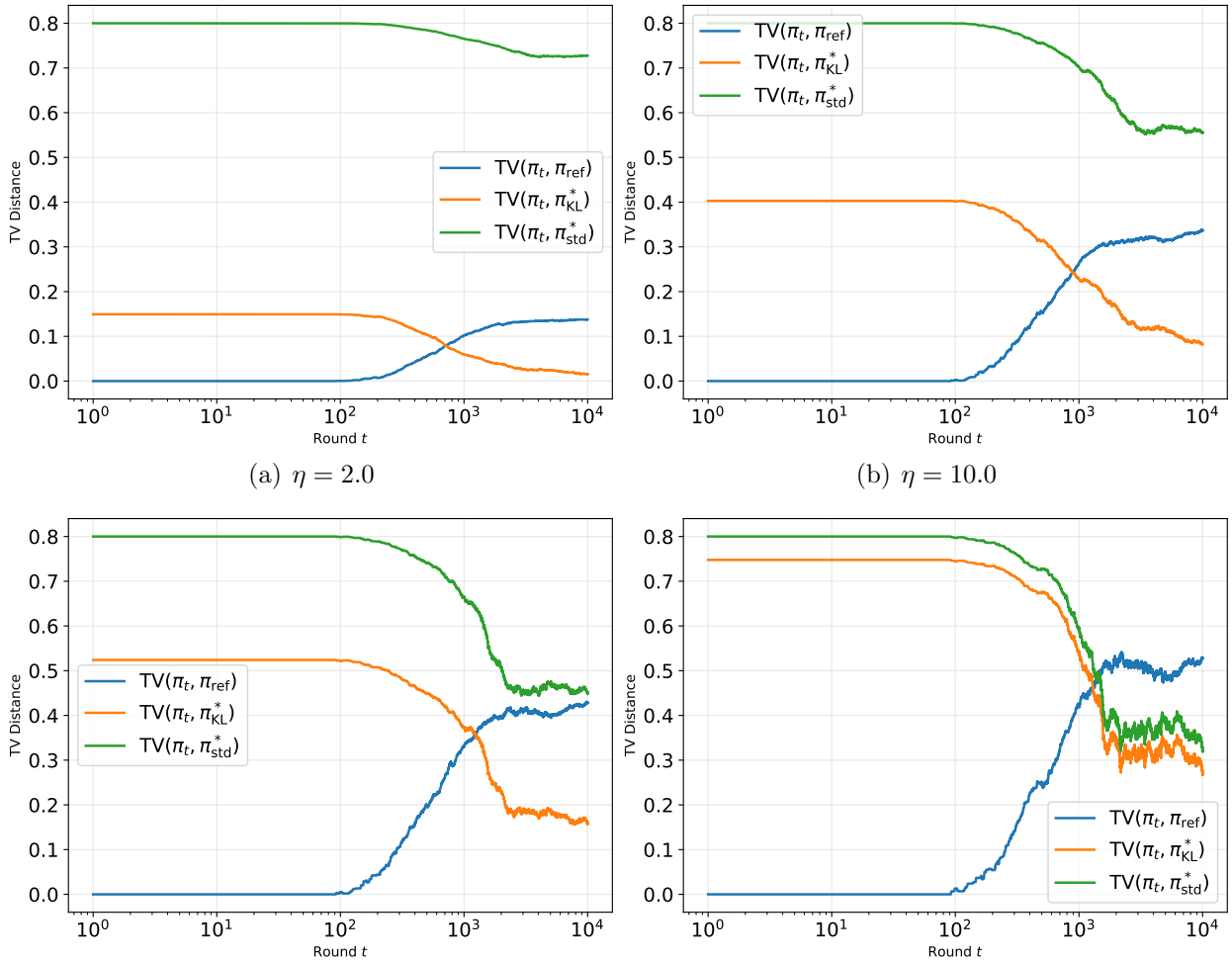



(a) $\eta = 5.0$ (b) $\eta = 20.0$

(c) $\eta = 100$ (d) $\eta = 500.0$

(e) $\eta = \infty$ (f) Comparison over $\eta$

Figure 1: The regret curves of KL-UCB for different values of $\eta$. In these experiments, we consider $\eta \in \{5, 20, 100, 500, \infty\}$, where $\eta = \infty$ corresponds to the standard MAB setting. We set $K = 4$ and $T = 10^6$. The bandit instances are defined by $\pi^{\mathrm{ref}} = \mathsf{Unif}(\mathcal{A})$ and rewards $r = [\delta, 2\delta, 0, 0]$, where $\delta$ is swept log-uniformly over an appropriate interval. For each fixed $\eta$, the regret corresponding to each instance (determined by $\delta$) is plotted as a single blue solid line in Figures 1(a)–1(e). The **red bold dashed lines** represent the maximum regret over all instances, and thus approximate the minimax regrets. For ease of comparison, all minimax regret curves are plotted together in Figure 1(f). **Both axes are displayed on a log scale.** We observe that the minimax regret curve for the standard MAB setting exhibits a slope of $1/2$ for $t > 100$, indicating a $\sqrt{T}$ rate. In contrast, when $\eta < \infty$, the slope of the regret curve decreases continuously as $t$ increases, with more pronounced decreases for smaller values of $\eta$. These results are consistent with the theoretical guarantees established in the paper.




(a) $\eta = 2.0$ (b) $\eta = 10.0$

(c) $\eta = 20.0$ (d) $\eta = 50.0$

Figure 2: Policy convergence in TV distance across training rounds $t \in [T]$ for varying regularization strengths $\eta \in \{2, 10, 20, 50\}$, with $K = 5$, $T = 10^4$, and $\pi^{\mathsf{ref}} = \mathsf{Unif}(\mathcal{A})$. Each subfigure plots the TV distance between the learned policy $\pi_t$ and three comparison policies: the reference policy $\pi_{\mathrm{ref}}$, the optimal policy $\pi^*_{\mathrm{KL}}$ for the KL-regularized regret, and the deterministic optimal policy $\pi^*_{\mathrm{std}}$ for the standard regret. For small $\eta$ (strong regularization), $\pi_t$ remains close to $\pi_{\mathrm{ref}}$ and converges toward $\pi^*_{\mathrm{KL}}$, consistent with the theoretical prediction that KL penalty dominating the objective. As $\eta$ increases, the regularization weakens and the policy drifts away from $\pi_{\mathrm{ref}}$, instead approaching $\pi^*_{\mathrm{std}}$, thereby recovering the unregularized optimum. This behavior illustrates how $\eta$ interpolates between the two regimes, complementing our theoretical guarantees on policy convergence.